Implementation and Experimental Evaluation of FastSpeech 2-Based Bilingual Text-to-Speech Synthesis for English and Telugu

Technical Report | IIIT Hyderabad – Language Technologies Research Centre Assignment

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A technical account of implementation, training evidence, inference outputs and variance-adaptor analysis.

Abstract

This report documents the implementation and experimental evaluation of two separate FastSpeech 2 text-to-speech systems for English and Telugu. The work adapts a reference FastSpeech 2 design into a shared PyTorch implementation with language-specific preprocessing, text handling, configuration, training and inference entry points. The English system uses the AI4Bharat Svarah corpus and the Telugu system uses IndicTTS Telugu resources and existing Telugu alignments. Both configurations specify 30,000 training steps and a common acoustic feature setup of 22.05 kHz audio, 80 mel channels, a 256-sample hop and an 1,024-point FFT. The report analyses recorded training logs, generated test utterances, and the available duration, pitch and energy variance-adaptor visualisations. The final inference outputs were generated with Griffin–Lim because the configured HiFi-GAN generator checkpoint was not available in the execution environment. This report therefore distinguishes verified system execution and numerical evidence from perceptual quality claims that were not formally measured.

Contents

1. Introduction and objectives
2. FastSpeech 2 background and architecture
3. Software design and repository organisation
4. Datasets and experimental protocol
5. Preprocessing and language-specific handling
6. Model configuration and training

7. Training and validation results
8. Inference and test-sentence results
9. Variance-adaptor analysis
10. Comparative discussion
11. Limitations and future work
12. Reproducibility and execution
13. Conclusion

1. Introduction and objectives

Text-to-speech synthesis

Text-to-speech (TTS) converts written language into a speech waveform. A practical TTS pipeline must represent text linguistically, predict a sequence of acoustic features, and convert those features into an audible signal. The assignment asks for separate English and Telugu FastSpeech 2 systems, trained with the specified corpora, and evaluated on the supplied test sentences.

Assignment objectives

The implementation target was not merely to call a pretrained synthesiser. The assignment required understanding and adapting a reference implementation, maintaining separate language models, documenting dataset selection and preprocessing, generating five test samples per language, and analysing the duration, pitch (F0) and energy outputs of the variance adaptor. The repository structure and the report are designed around those requirements.

Scope of this report

The report describes the code and experimental artifacts present in the supplied project archive. It treats the archived configuration, logs, inference summaries, audio files and plots as primary evidence. Where the archive does not contain a verifiable statistic or listening assessment, the report explicitly avoids inventing one.

2. FastSpeech 2 background and architecture

Architecture overview

FastSpeech 2 is a non-autoregressive acoustic model. Rather than generating one acoustic frame at a time conditioned on all previously generated frames, it uses a duration expansion mechanism to align encoded linguistic units with acoustic-frame sequences. This enables parallel acoustic prediction during inference and gives the model explicit control over prosodic factors.

Text representation and encoder

The input text is converted to language-appropriate units. The English path uses phoneme-oriented processing, while the Telugu path uses Unicode grapheme/code-point units as described by the project implementation. The encoder transforms the unit sequence into contextual hidden representations using FFT-style blocks.

Variance adaptor

The variance adaptor predicts duration, pitch and energy information. Duration prediction determines how long each input unit is represented in the expanded sequence. Pitch and energy predictors provide prosodic conditioning; their embeddings are combined with the expanded encoder representation before decoding. The project produces per-sentence plots for all three predicted quantities.

Decoder and acoustic output

The decoder maps the expanded and variance-conditioned sequence to mel-spectrogram frames. A postnet refines the mel prediction. The waveform-generation stage then converts the mel representation to audio. In the final archived

inference, HiFi-GAN could not be loaded because its generator checkpoint was absent; Griffin–Lim was used as a fallback. Thus the acoustic model and waveform reconstruction stages should be evaluated separately.

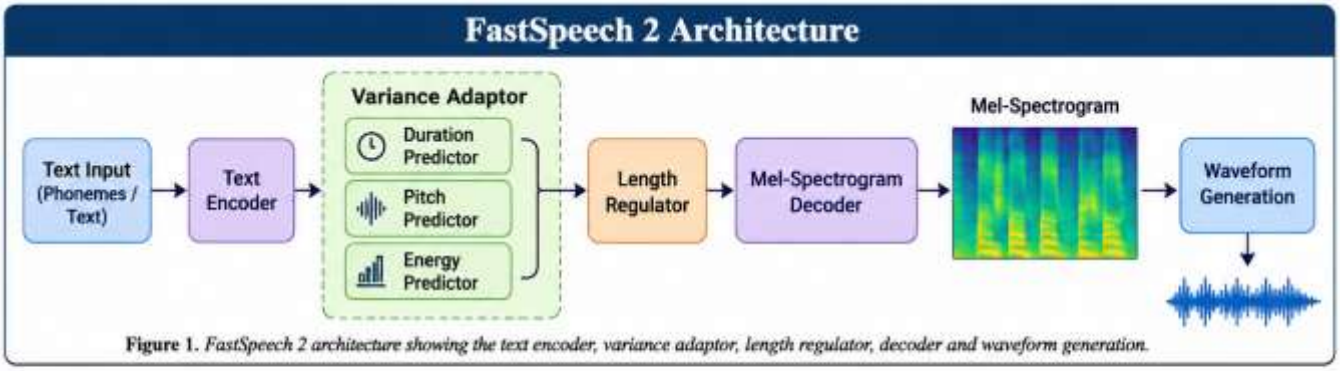
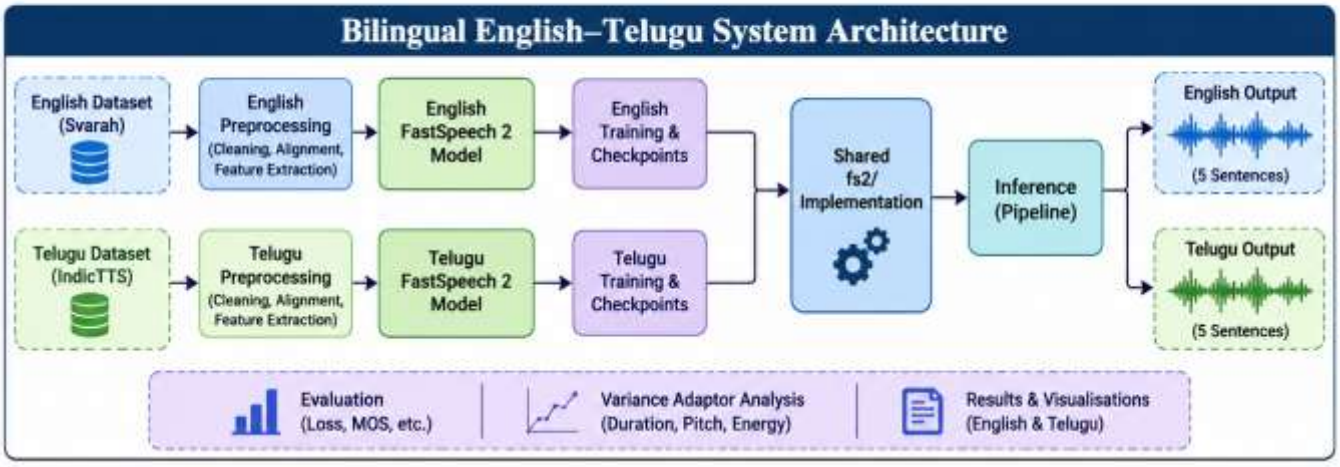


Figure 1. *FastSpeech 2 architecture showing the text encoder, variance adaptor, length regulator, decoder and waveform generation.*



3. Software design and repository organisation

Shared implementation with language-specific interfaces

The repository separates common model and pipeline code under `fs2/` from the language-specific folders. The English and Telugu `train.py`, `preprocess.py` and `inference.py` files are compact command entry points that delegate to shared implementation modules. Their brevity is a modularity decision: common training and inference logic is maintained once, while language-specific configuration and text handling remain separate.

Core modules

`fs2/model.py` defines the model components; `fs2/train_lib.py` contains training functionality; `fs2/infer_lib.py` implements inference; `fs2/pipeline.py` coordinates data preparation, feature extraction and stages; `fs2/data.py` handles dataset access; `fs2/audio.py` provides audio/feature utilities; `fs2/text_en.py` and `fs2/text_te.py` implement language-specific text processing; `fs2/textgrid.py` handles alignment data; and `fs2/cli.py` exposes command-line entry points.

Reproducibility structure

The `English/` and `Telugu/` directories contain their own configuration files and test sentence JSON files. `Results/English/` and `Results/Telugu/` contain loss plots, loss logs, inference summaries, generated WAV files and variance-adaptor plots. The project archive used to prepare this report did not contain the trained `.pt` checkpoint binaries; the checkpoint files are stored separately in the working Drive project and must be made accessible in the final repository using Git LFS or a persistent external download link.

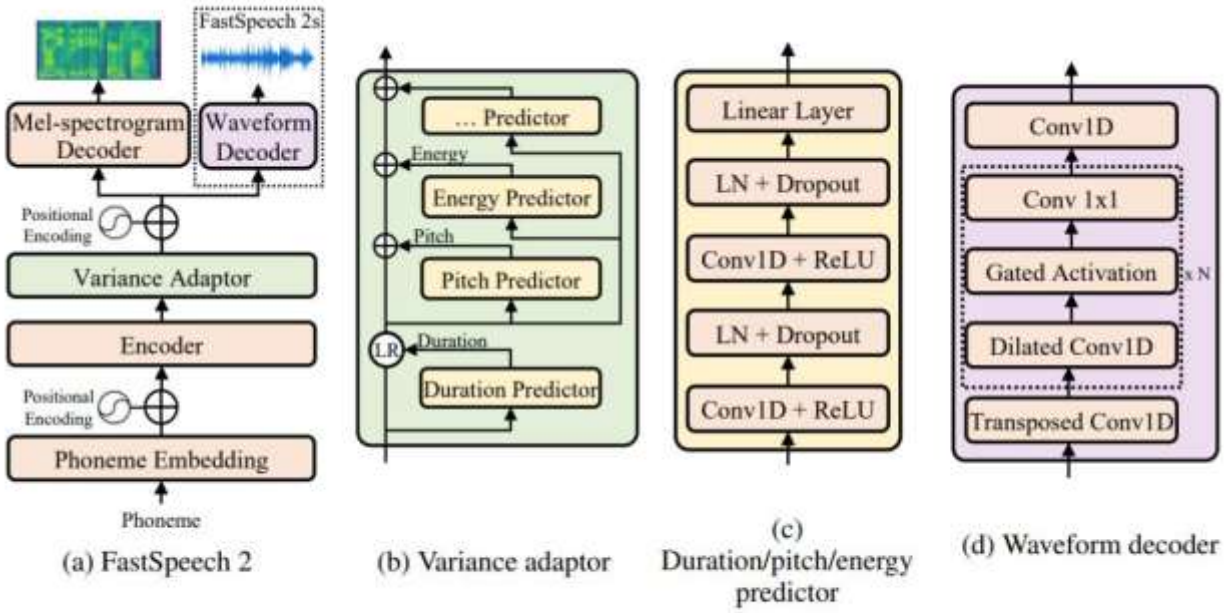


Figure 1: The overall architecture for FastSpeech 2 and 2s. LR in subfigure (b) denotes the length regulator proposed in FastSpeech. LN in subfigure (c) denotes layer normalization.

4. Datasets and experimental protocol

English: Svarah

The recorded English workflow uses AI4Bharat Svarah. The configuration specifies a target of 5.0 hours for training and 1.0 hour for validation, a duration filter of 1.5 to 15.0 seconds, and seed 1234. The recorded audit reports 4,377 downloaded WAV files at 16 kHz, with zero missing files and no duplicate IDs. The split audit records 3,140 training recordings and 610 validation recordings, with no recording-ID overlap. The feature-extraction record reports 3,138 training segments and 610 validation segments; two training segments were excluded as `spn`. The recorded vocabulary size is 71.

Telugu: IndicTTS Telugu

The Telugu configuration points to the recovered Telugu manifests, a Telugu alignment corpus, existing alignment TextGrids, and a grapheme-to-grapheme dictionary. The implementation uses existing alignments rather than re-running the alignment stage. The configuration splits longer utterances at sufficiently long silences so that resulting segments do not exceed 15 seconds. The supplied archive does not include a complete numeric Telugu duration/split audit comparable to the English audit; this report therefore does not assign an unsupported Telugu hour count or utterance count.

Split integrity and scope

For English, the recorded split audit explicitly establishes no recording-ID overlap. For Telugu, the configuration references distinct training and validation manifests, but the ZIP does not contain a complete split audit proving every underlying source recording is disjoint. The report describes the separate manifests and avoids extending the English audit claim to Telugu without equivalent evidence.

5. Preprocessing and language-specific handling

Audio and acoustic features

Both language configurations specify a 22,050 Hz model sample rate, FFT size 1,024, hop size 256, window size 1,024, 80 mel channels, and a mel frequency range from 0 to 8,000 Hz. The README describes mel features, F0 extraction

with PyWorld, energy derived from an STFT magnitude norm, and training-statistics normalisation. The exact processing implementation is shared where possible and parameterised by language configuration.

English text and alignment

The English pipeline uses Montreal Forced Aligner with the `english_us_arpa` acoustic model. The archived workflow logs report 3,750 TextGrids. The English text module includes a pronunciation dictionary path and a stress-related dictionary correction. The feature log records two training segments dropped as `spn`, leaving 3,138 training and 610 validation feature segments.

Telugu text processing

The Telugu path is designed around Telugu grapheme/code-point units and a grapheme-to-grapheme dictionary. The README records that whitespace and punctuation are not treated as ordinary input tokens, and punctuation symbols are mapped to a pause token. The pipeline uses existing Telugu TextGrids and audits alignment coverage during preparation. Long utterances are segmented at aligned silences, with a maximum segment duration of 15 seconds.

Why separate language paths matter

English phoneme conversion and Telugu grapheme processing are not interchangeable. Keeping separate text modules, dictionaries, manifests and configurations allows the acoustic model and training pipeline to share a common structure without forcing the two scripts into the same linguistic representation.

6. Model configuration and training

Common acoustic configuration

The two YAML files specify model dimension 256, four encoder layers, four decoder layers, two attention heads, feed-forward dimension 1,024, kernel size 9, dropout 0.2, variance-adaptor filter dimension 256, variance kernel size 3, and 256 quantisation bins. The audio feature configuration is common across the two languages.

Training schedule

Both configurations specify 30,000 training steps, 4,000 warm-up steps, learning rate 0.001, gradient clipping at 1.0, logging every 100 steps, validation every 1,000 steps, checkpoint saving every 2,000 steps, batch frame budget 14,000, maximum batch size 32, and seed 1234. These are configuration values; they should not be confused with an independent re-run performed for this report.

Recorded validation results

The project facts file records best validation loss 3.6364187819058778 for English and 2.183233747507799 for Telugu. These values are reported as recorded project results. Loss values are dependent on implementation, data representation and objective scaling; the numerical difference between languages should not be interpreted as a direct ranking of speech quality.

7. Training and validation results

How to interpret the curves

The training and validation curves should be read together. A declining training loss indicates that the model is fitting the training objective. Validation loss provides a check on behaviour on held-out data. A persistent gap or rising validation loss can be consistent with overfitting, but loss alone does not establish perceptual intelligibility or naturalness.

English log

The archived English loss CSV contains 350 rows (319 training records and 31 validation records). Its recorded training total loss begins at 13.2138 and ends at 1.0701; the minimum recorded training total is 1.0700. The first recorded validation total is 4.1928, the final recorded validation total is 3.7010, and the minimum logged validation total is 3.6364 at step 13000. The report includes the saved training and validation plots as the visual record.

Telugu log

The archived Telugu loss CSV contains 357 rows (325 training records and 32 validation records). Its recorded training total loss begins at 13.9902 and ends at 0.8980; the minimum recorded training total is 0.8968. The first recorded validation total is 2.6989, the final recorded validation total is 2.1979, and the minimum logged validation total is 2.1832 at step 11000.

Interpretation and caution

The saved plots and CSVs provide evidence of optimisation progress. Validation values fluctuate between evaluation points, which is expected in finite-data neural training. The final validation value need not equal the best value. The recorded best validation values are therefore stated separately from the last validation values. No claim of absence of overfitting is made solely from the existence of a decreasing training curve.

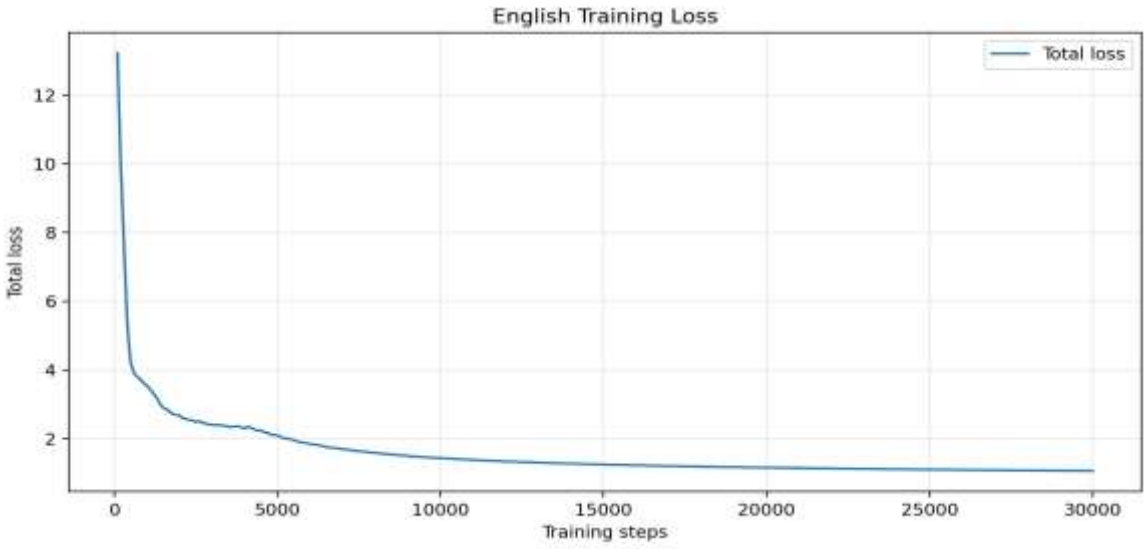


Figure: English training loss over recorded optimisation steps.

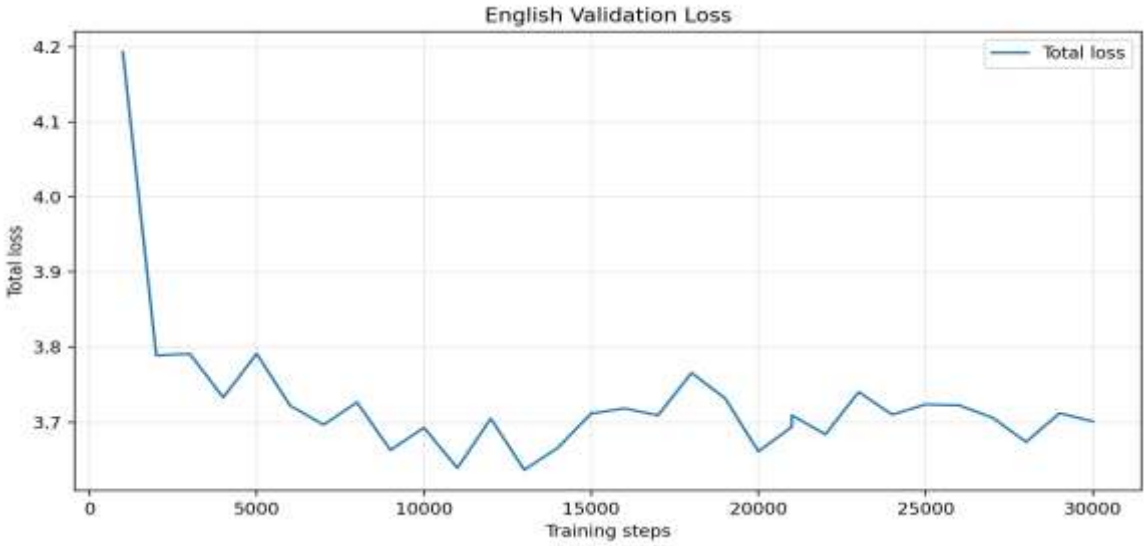


Figure: English validation loss at recorded evaluation steps.

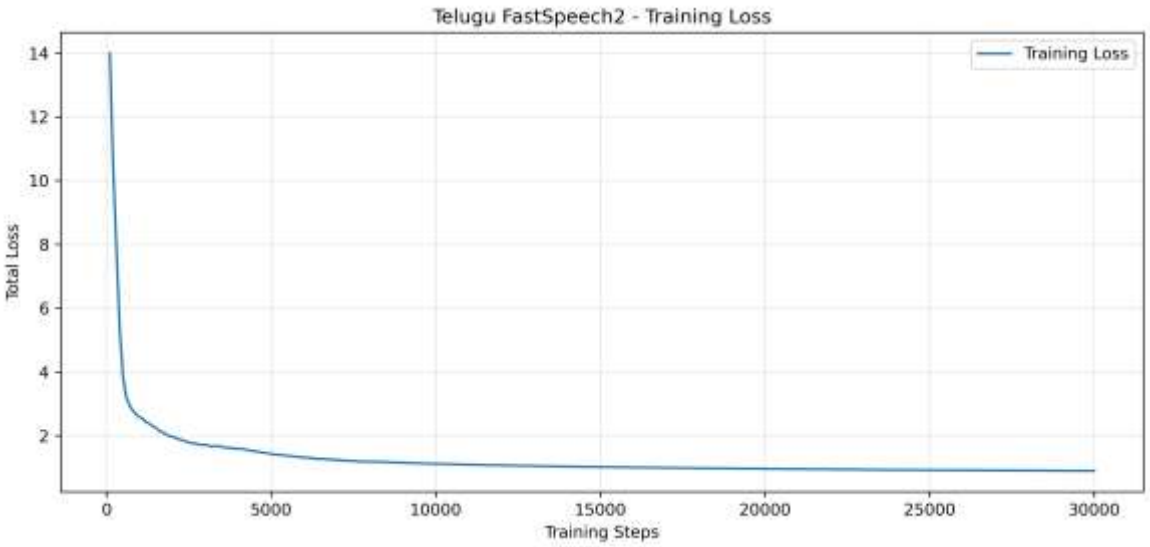


Figure: Telugu training loss over recorded optimisation steps.

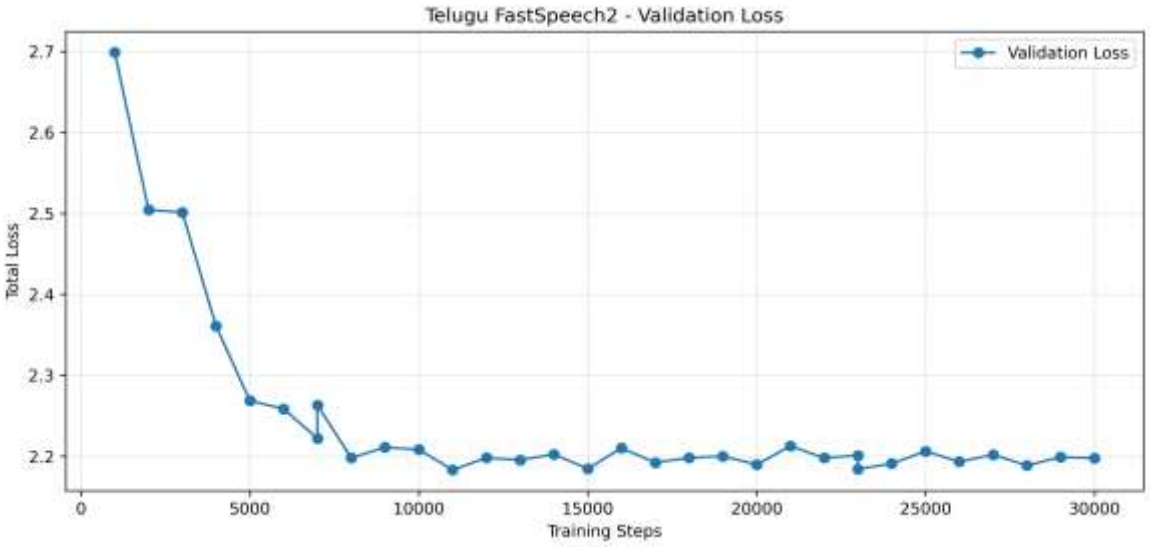


Figure: Telugu validation loss at recorded evaluation steps.

Language	Log rows	Train first	Train last	Best val	Best-val step
English	350	13.214	1.070	3.636	13000
Telugu	357	13.990	0.898	2.183	11000

8. Inference and test-sentence results

Test protocol

The assignment asks for five test samples per language and says to preserve the supplied sentence text. The project stores the exact test strings in language-specific JSON files and records the generated waveform outputs and inference summaries. The English E3 source string is reproduced exactly as printed in the assignment PDF, including the internal space in “rhyth m”; this appears to be a typographical anomaly in the assignment source and was not silently corrected.

Audio inventory

The generated files are WAV audio at 22,050 Hz. The table below lists the durations measured from the WAV headers in the supplied archive. Duration and sample rate verify that the files were generated; they do not, by themselves, establish pronunciation correctness or listener-rated quality.

Per-sentence evaluation

The inference summaries include normalised text and predicted phone/grapheme sequences, duration-frame arrays, pitch arrays and energy arrays. These make it possible to trace the acoustic predictions for each sentence. The plots are analysed as model outputs, not as a substitute for a formal listening study. No MOS, ABX, word-error or listener-intelligibility score is present in the archive.

Language	ID	WAV	Sample rate	Duration (s)
English	E1	E1.wav	22,050	6.769
English	E2a	E2a.wav	22,050	4.272
English	E2b	E2b.wav	22,050	4.319
English	E3	E3.wav	22,050	3.576
English	E4	E4.wav	22,050	7.686
Telugu	T1	T1.wav	22,050	4.899
Telugu	T2a	T2a.wav	22,050	2.403
Telugu	T2b	T2b.wav	22,050	2.705
Telugu	T3	T3.wav	22,050	2.310
Telugu	T4	T4.wav	22,050	8.104

English official test sentences

E1. The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

E2a. The orchestra's conduct during the ceremony was praised by the audience.

E2b. The researchers will conduct an experiment to evaluate the proposed method.

E3. The rhyth m of the orchestra changed unexpectedly.

E4. The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Telugu official test sentences

T1. ■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■ ■■■■■ ■■■■■■ ■■■■■■■■■■■■ ■■■■■■■■■■■■■■■.

T2a. ■■■■■■■ ■■■■■ ■■■■■■ ■■■■■ ■■■■■■ ■■■■■.

T2b. ■■■ ■■■■■■ ■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■■.

T3. ■■■■■ ■■■■■■■ ■■■■■■■■■■■■ ■■■■■■■■■.

T4. ■ ■■■■■■■■ 2026 ■■■■■■■ 12■ ■■■■■ ■■■■■ 10:45 ■■■■■■■ ■■■■■■■■■■■■■■■■■, ■■■■■■■■■ ■■■■■■■■■ ■■■■■■■.

9. Variance-adaptor analysis

Duration

Duration predictions determine the number of acoustic frames allocated to each input unit. The per-sentence duration figures should be read alongside sentence length and linguistic segmentation. Very long final pause allocations or unusually concentrated durations may merit inspection, but they should not automatically be labelled errors without alignment and audio evidence.

Pitch (F0)

Pitch predictions represent the model's learned fundamental-frequency conditioning. The pitch plots show variation across the unit sequence. They can be used to inspect whether the model predicts a changing prosodic contour, but they do not independently prove natural intonation. Unvoiced regions and the representation used for unvoiced frames must be considered when interpreting the axes.

Energy

Energy predictions represent relative acoustic intensity conditioning. The plots show how the model distributes predicted energy across units. These values are model features and depend on the extraction and normalisation procedure; they should not be equated directly with calibrated sound-pressure level.

Sentence-level coverage

The archive contains duration, pitch and energy plots for E1, E2a, E2b, E3 and E4 in English and T1, T2a, T2b, T3 and T4 in Telugu. All available plots are included in the appendices with language and sentence identifiers. This gives the evaluator a direct visual record of all 30 requested variance-adaptor outputs.

English: variance-adaptor plots

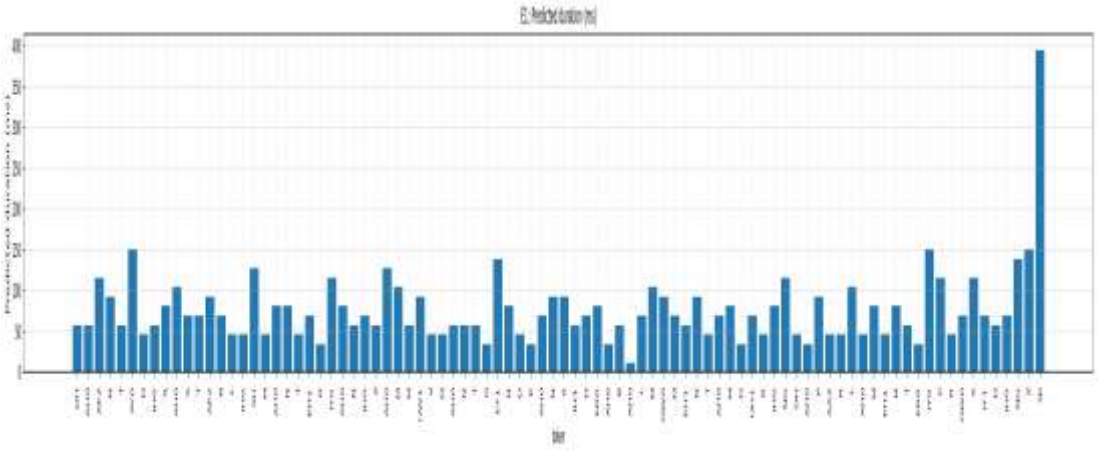


Figure: English E1 — predicted duration.

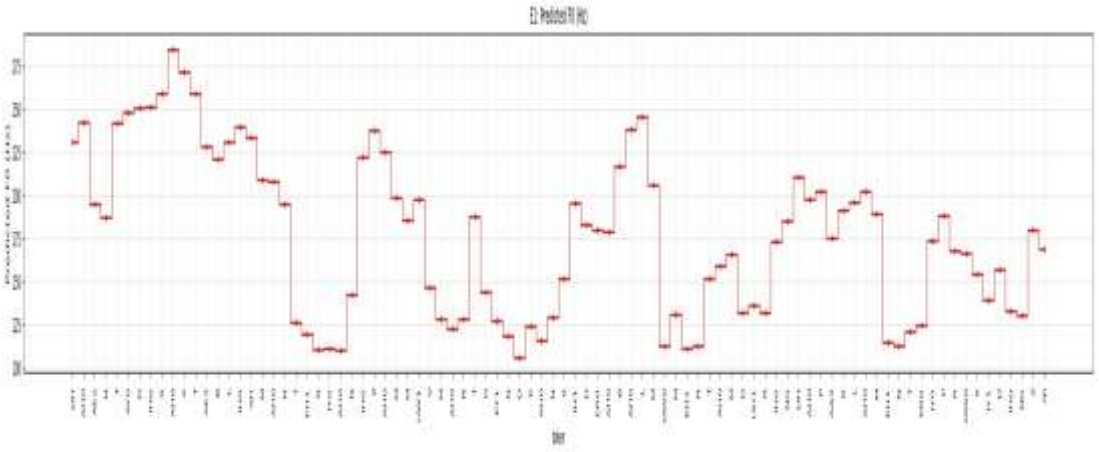


Figure: English E1 — predicted pitch.

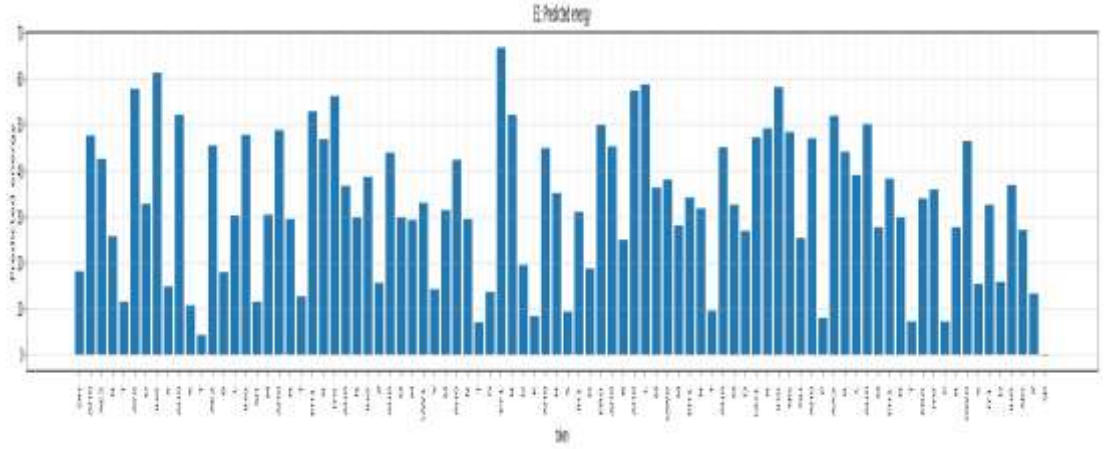


Figure: English E1 — predicted energy.

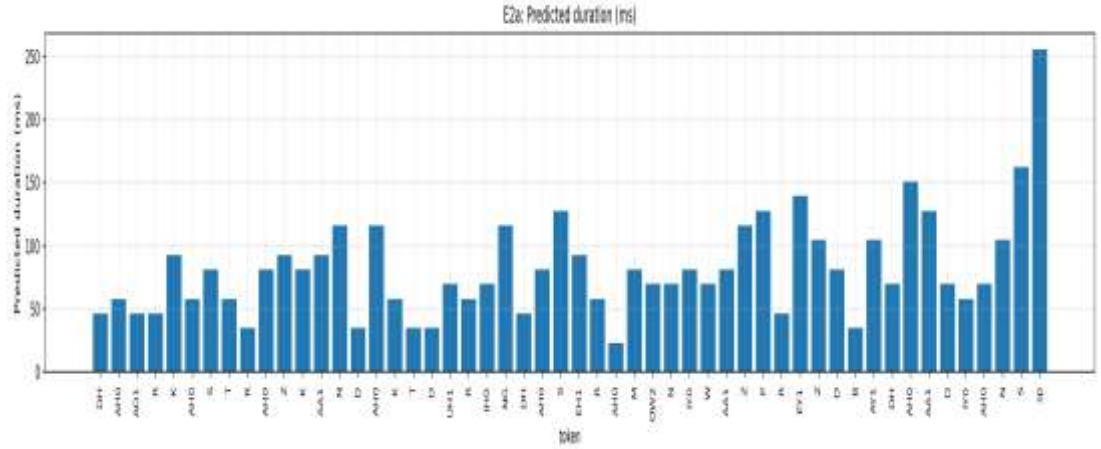


Figure: English E2a — predicted duration.

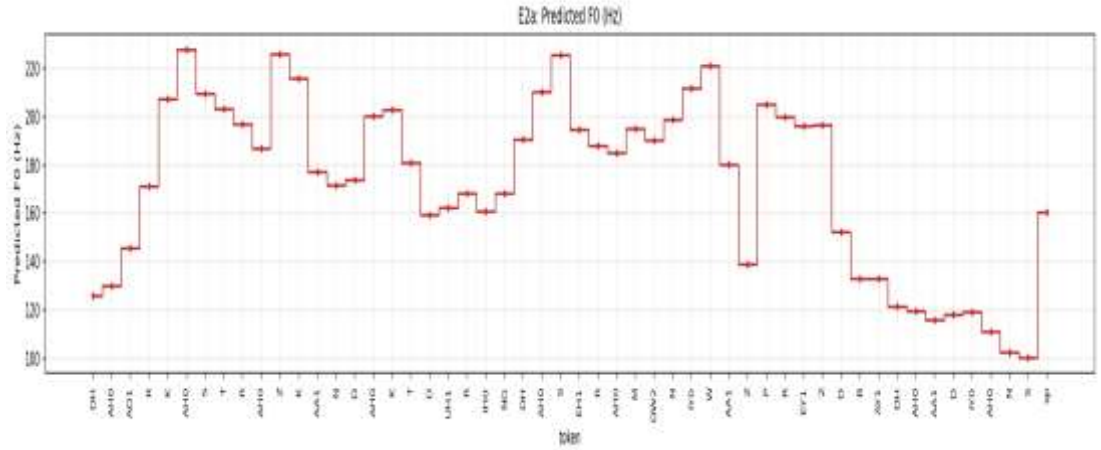


Figure: English E2a — predicted pitch.

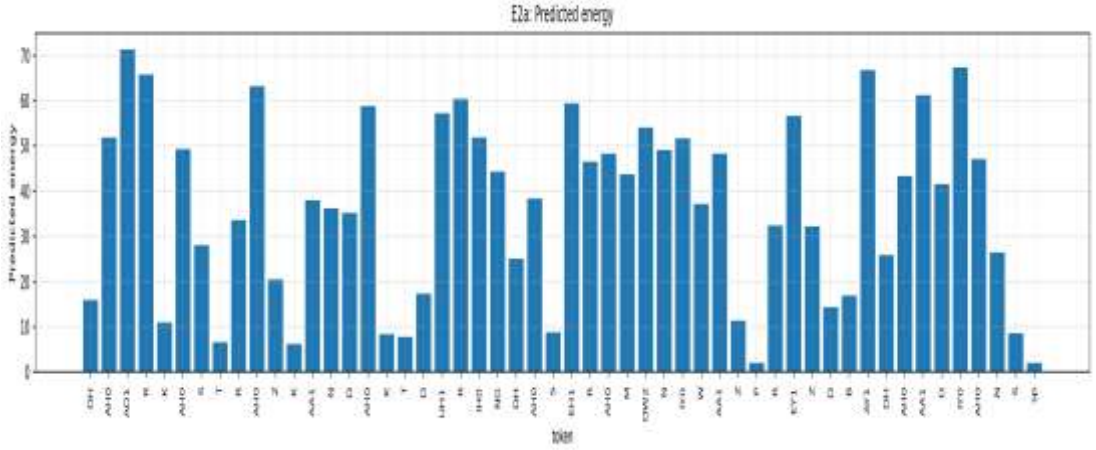


Figure: English E2a — predicted energy.

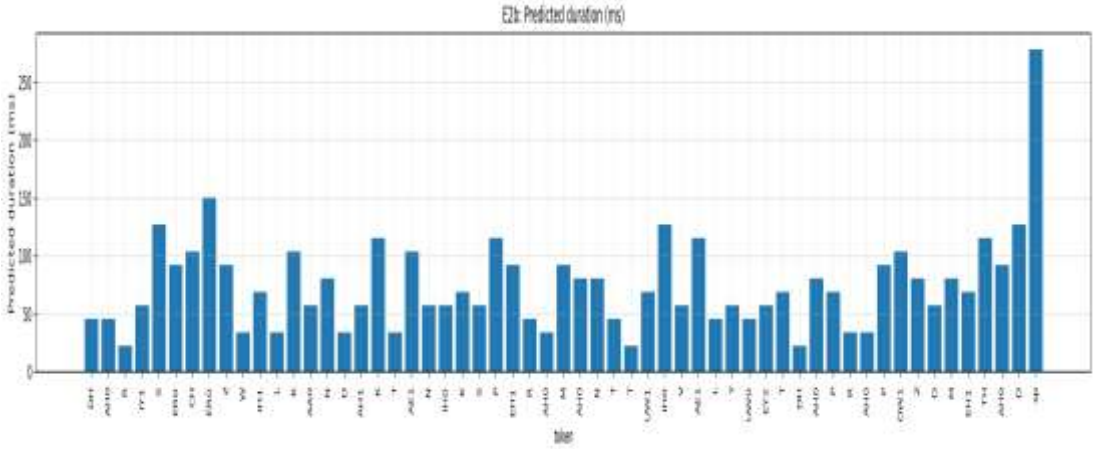


Figure: English E2b — predicted duration.

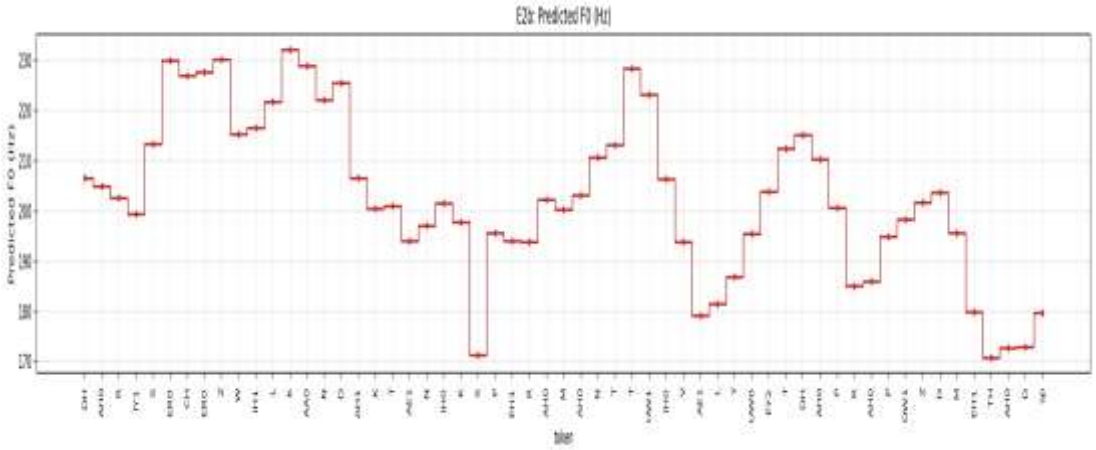


Figure: English E2b — predicted pitch.

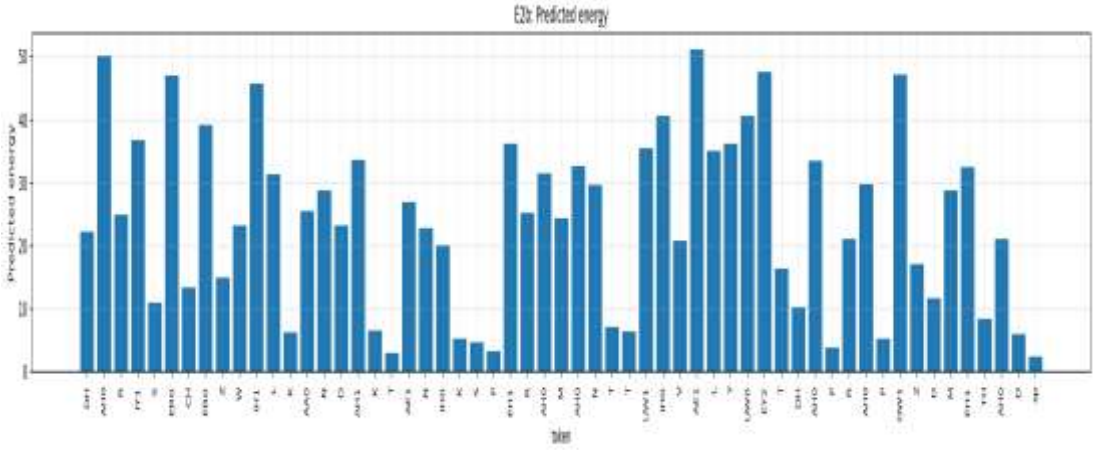


Figure: English E2b — predicted energy.

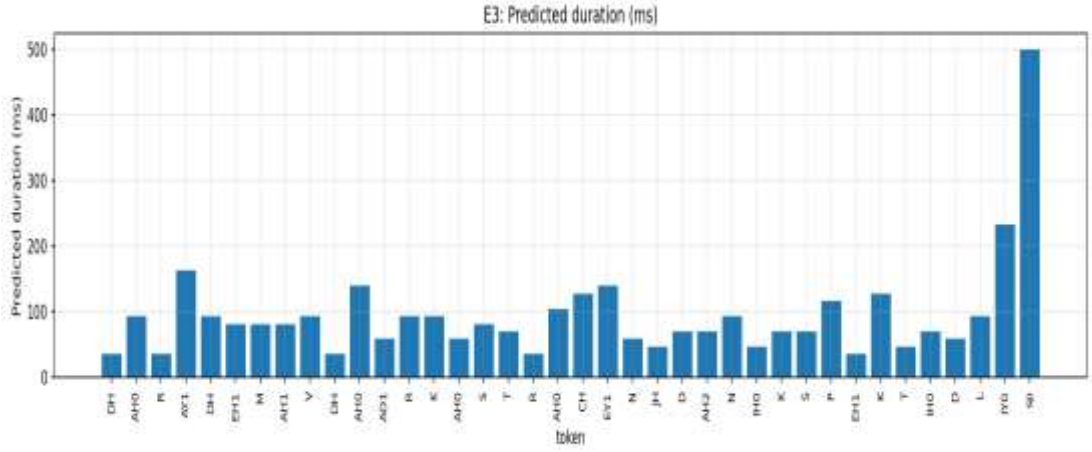


Figure: English E3 — predicted duration.

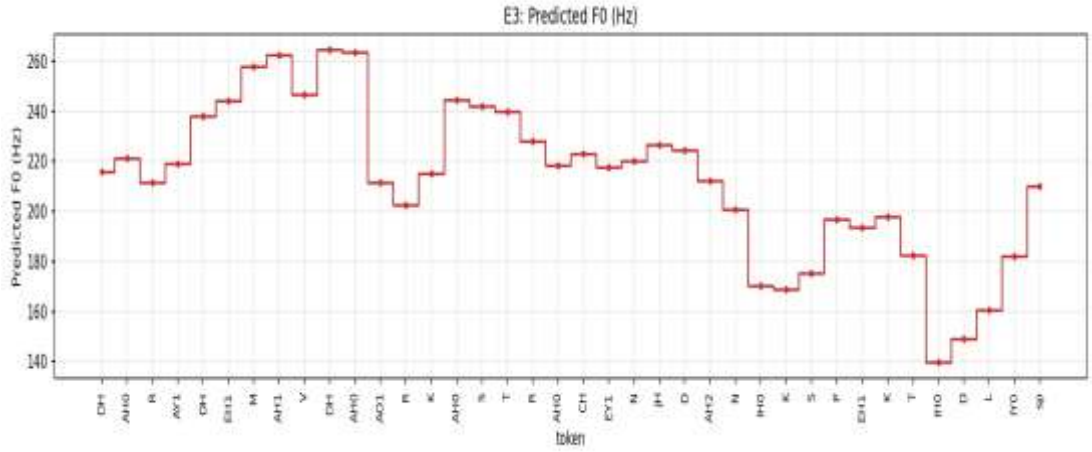


Figure: English E3 — predicted pitch.

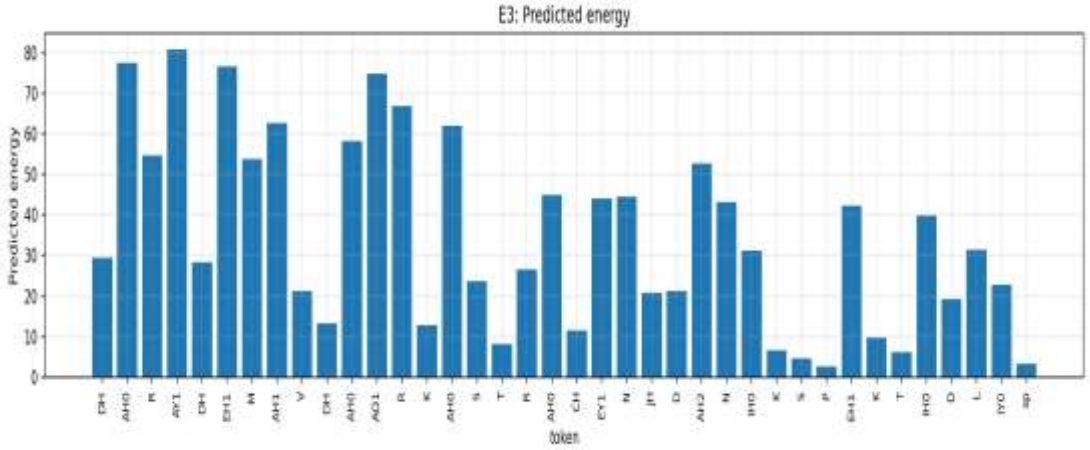


Figure: English E3 — predicted energy.

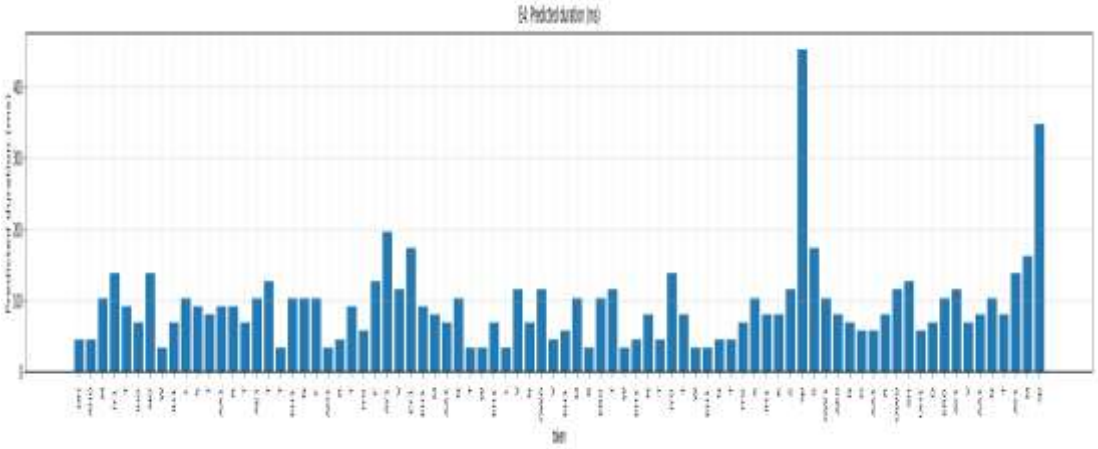


Figure: English E4 — predicted duration.

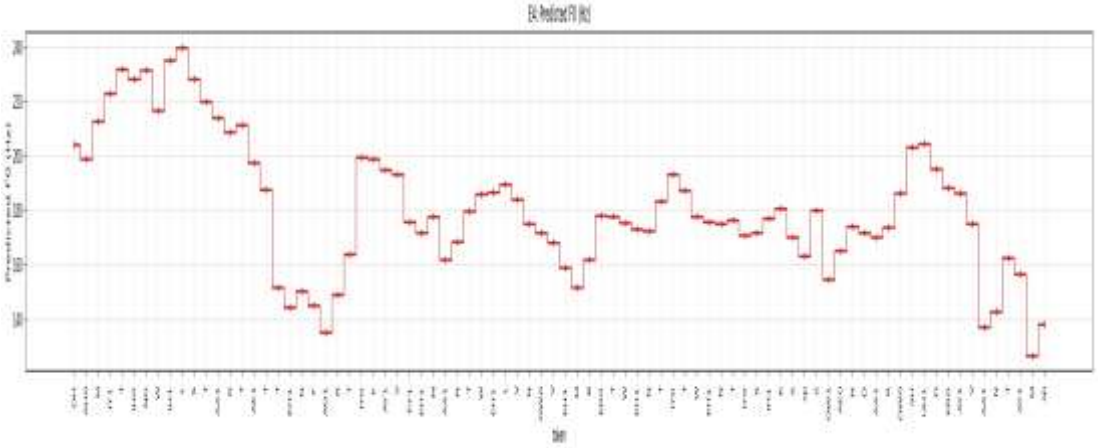


Figure: English E4 — predicted pitch.

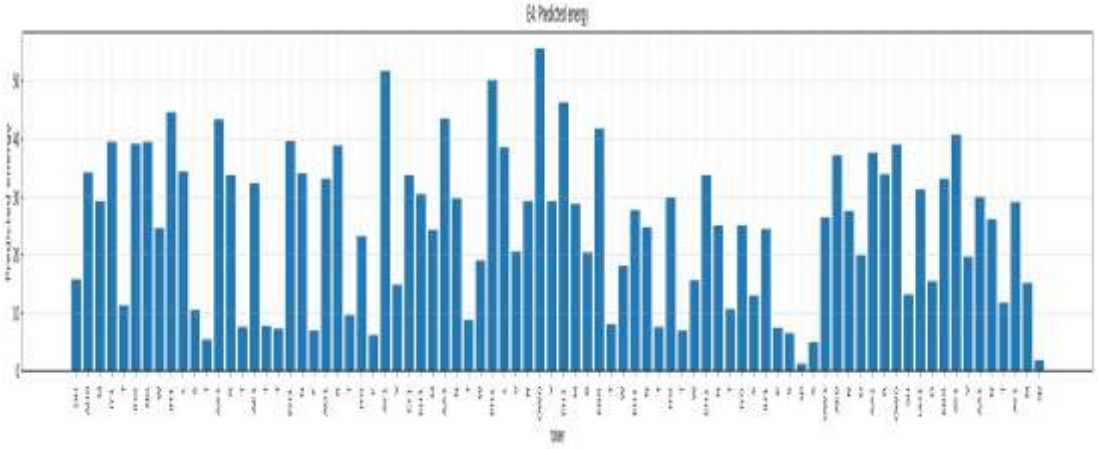


Figure: English E4 — predicted energy.

Telugu: variance-adaptor plots

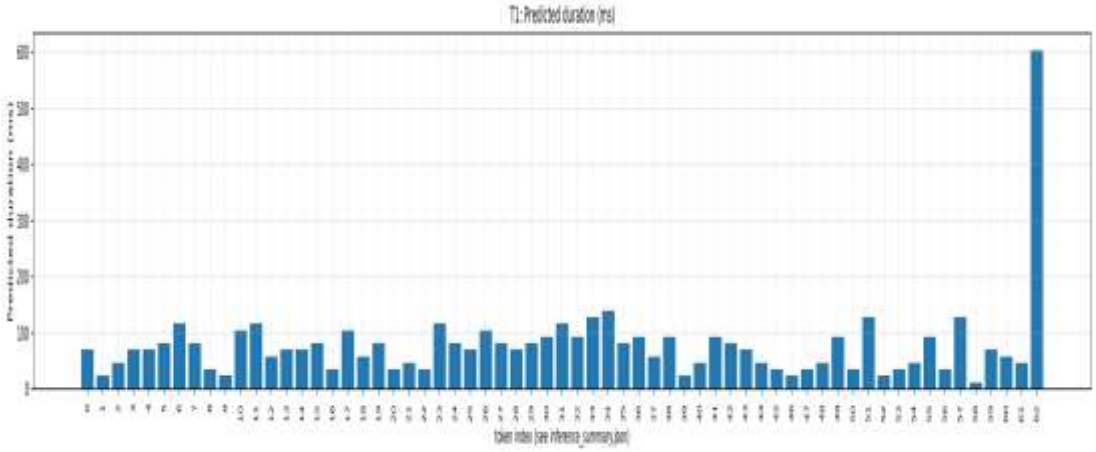


Figure: Telugu T1 — predicted duration.

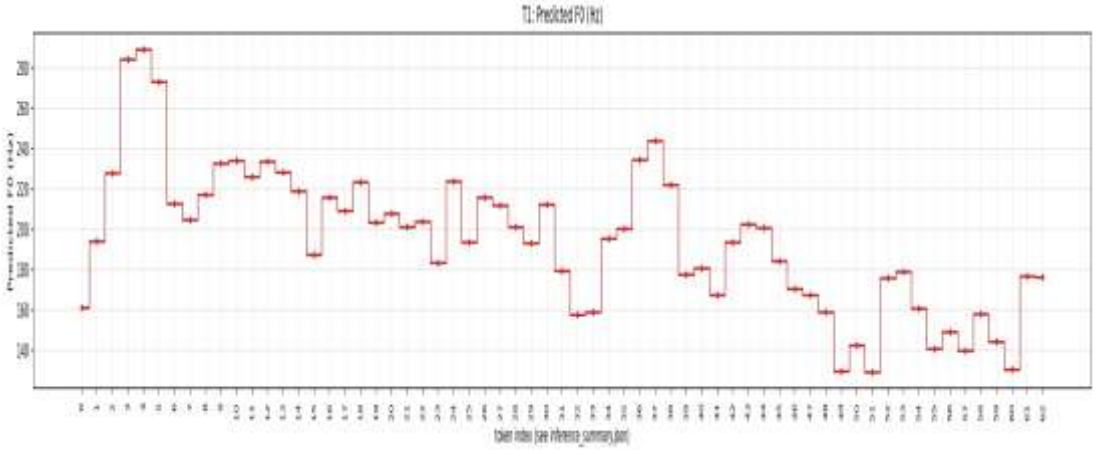


Figure: Telugu T1 — predicted pitch.

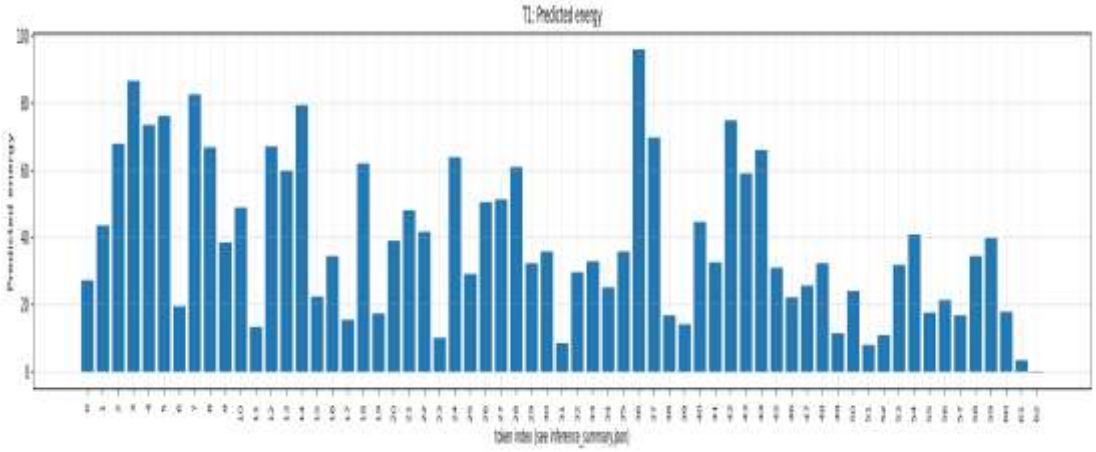


Figure: Telugu T1 — predicted energy.

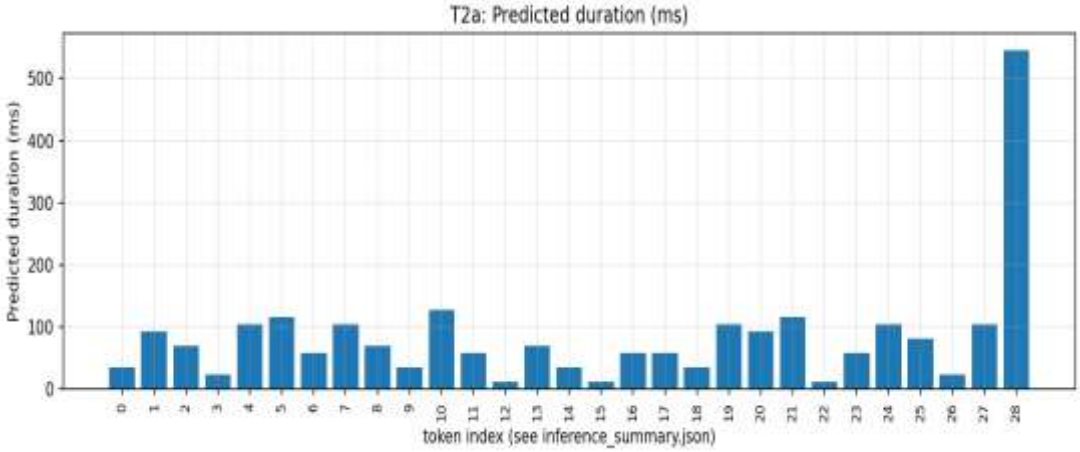


Figure: Telugu T2a — predicted duration.

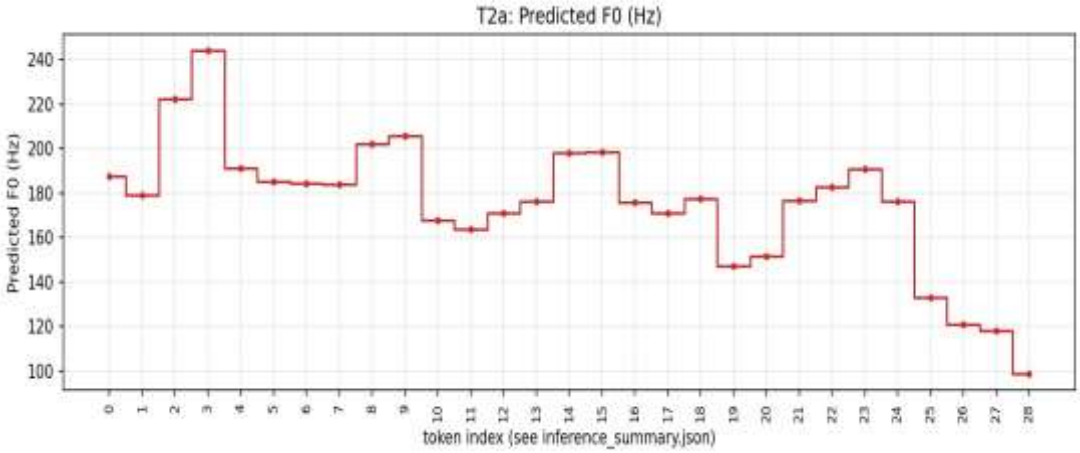


Figure: Telugu T2a — predicted pitch.

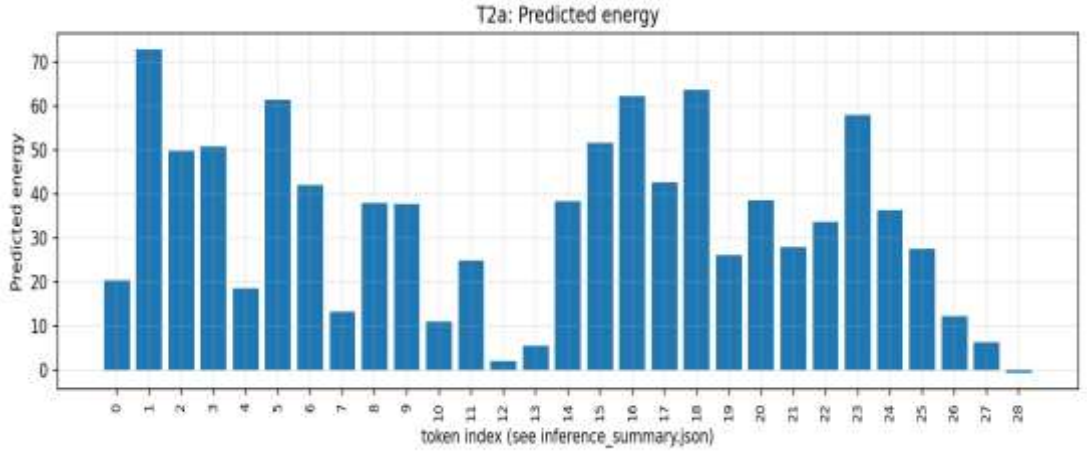


Figure: Telugu T2a — predicted energy.

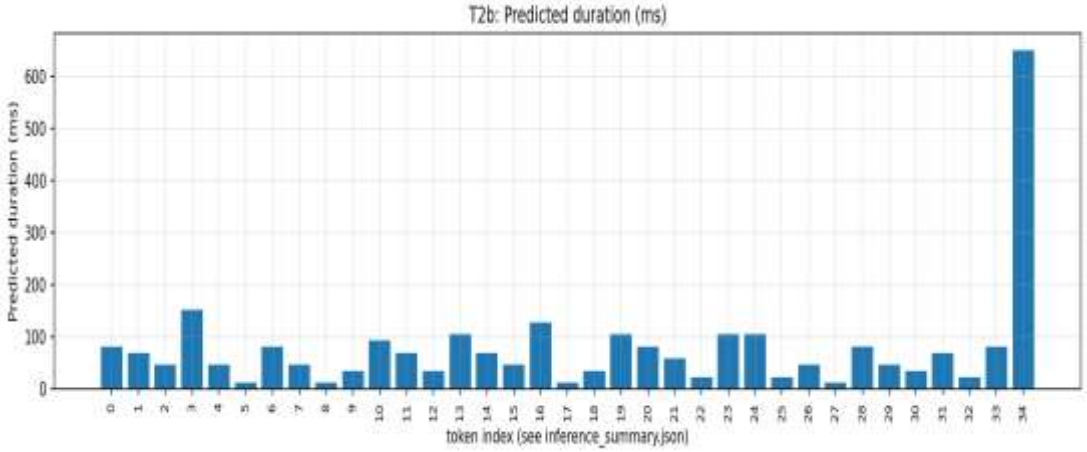


Figure: Telugu T2b — predicted duration.

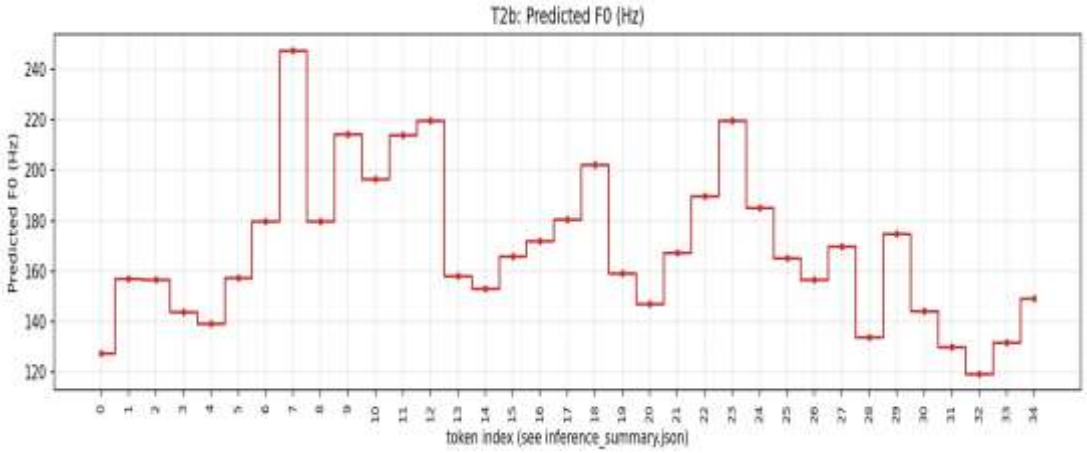


Figure: Telugu T2b — predicted pitch.

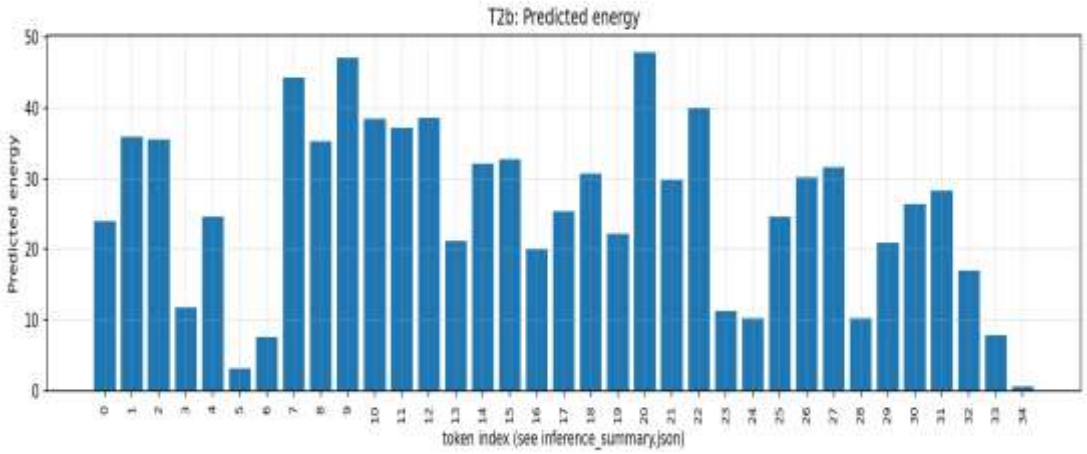


Figure: Telugu T2b — predicted energy.

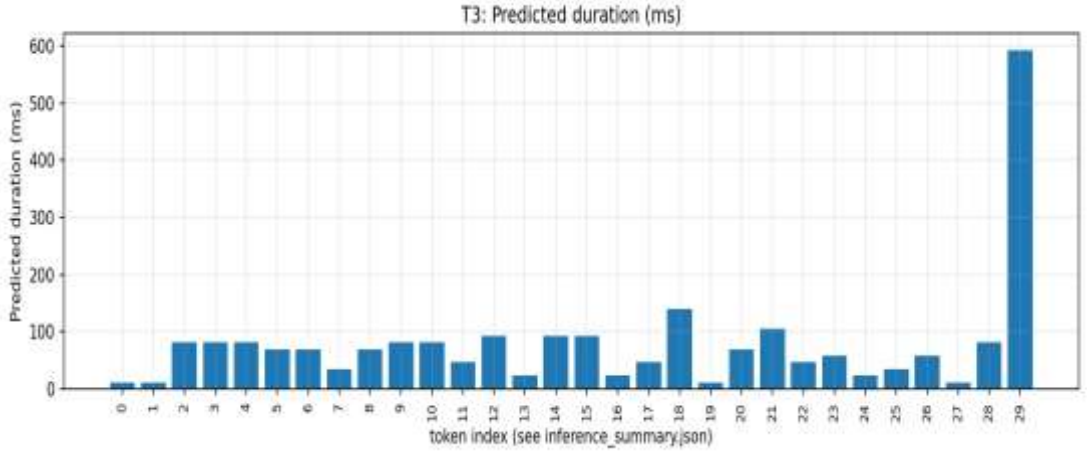


Figure: Telugu T3 — predicted duration.

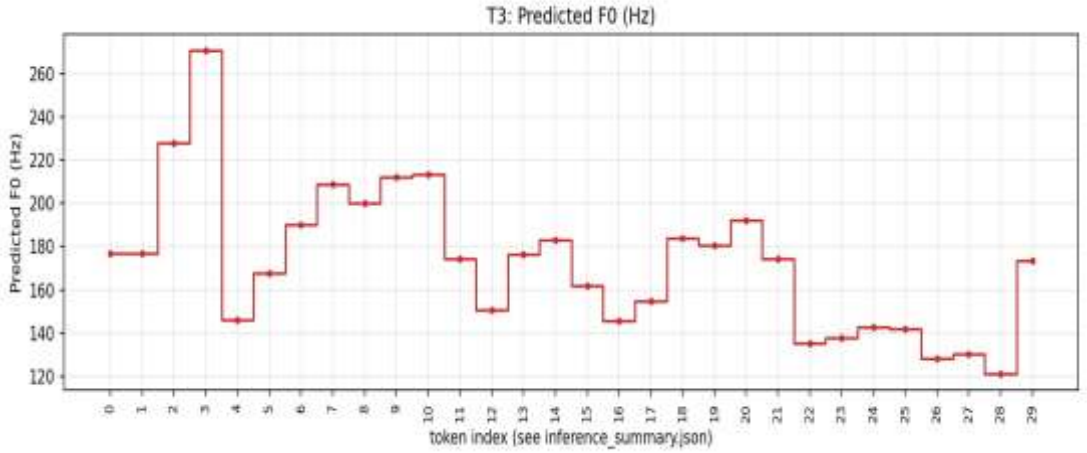


Figure: Telugu T3 — predicted pitch.

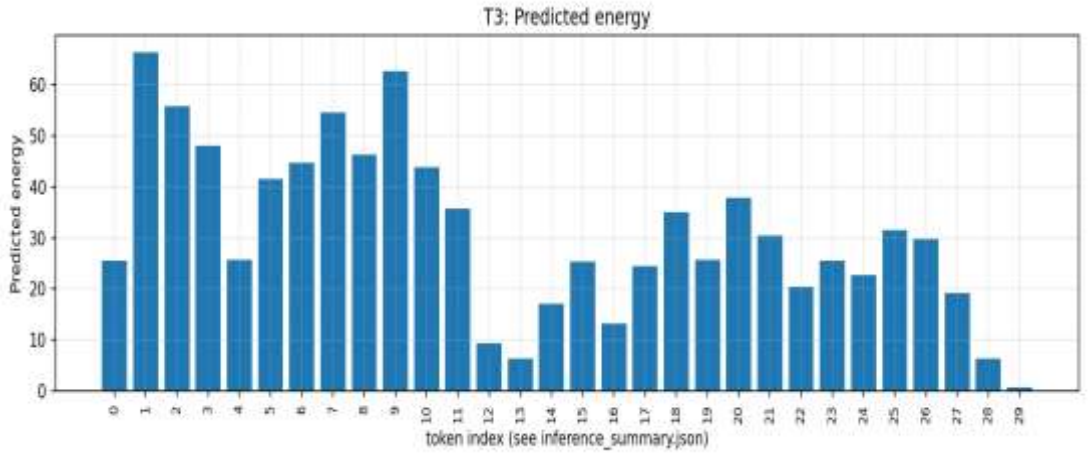


Figure: Telugu T3 — predicted energy.

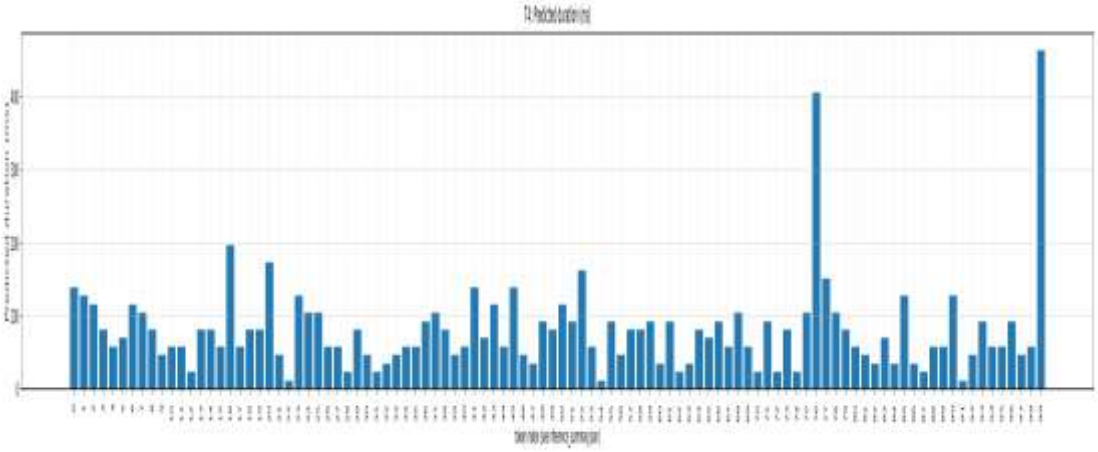


Figure: Telugu T4 — predicted duration.

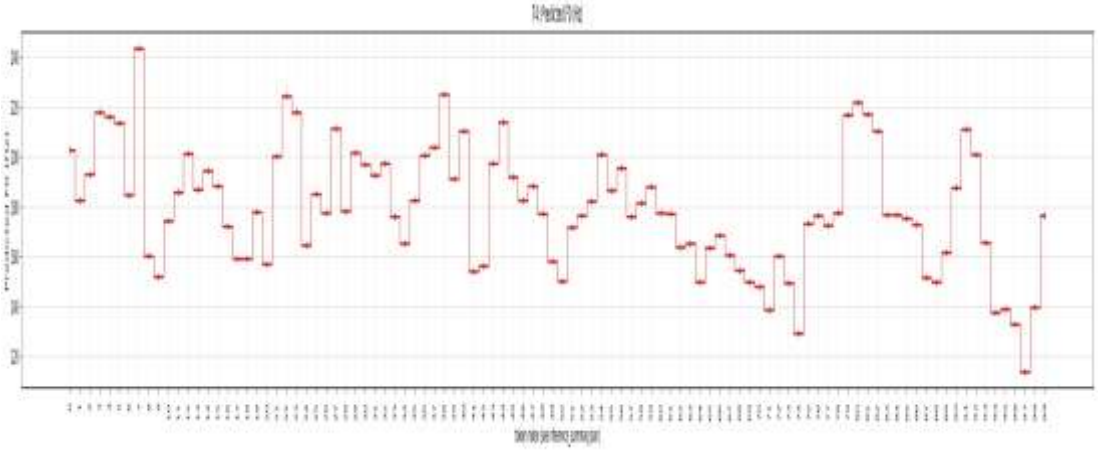


Figure: Telugu T4 — predicted pitch.

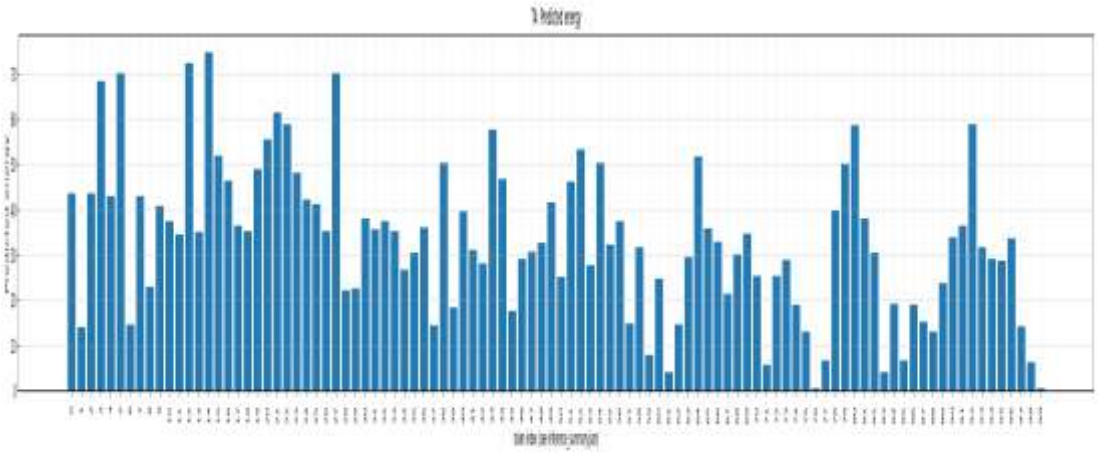


Figure: Telugu T4 — predicted energy.

10. Comparative discussion

Shared design, different linguistic units

The two systems share the same broad acoustic architecture and feature dimensions, while differing in text representation and data preparation. English uses phoneme-oriented processing and forced alignment with an English acoustic model; Telugu uses grapheme/code-point units and existing Telugu alignments. This separation is central to the bilingual implementation.

Loss comparison

The best recorded validation loss is lower for Telugu than English in the logged objective. However, because language-specific unit inventories, feature distributions and data pipelines can affect the scale and difficulty of the losses, the raw values are not a valid standalone comparison of synthesis quality.

Inference comparison

The audio inventory confirms five outputs for each language, all at the configured 22.05 kHz sample rate. Durations vary with sentence length and the model's predicted durations. The test set includes long or linguistically challenging material, including a very long English word, the contrastive use of “conduct”, a source-text spacing anomaly, a mixed-language name, Telugu grapheme sequences, and a time/date expression.

11. Limitations and future work

Waveform reconstruction

The HiFi-GAN generator checkpoint configured in the YAML files was not available at inference time. The pipeline therefore used Griffin–Lim. This allowed waveform generation and completion of the sample inventory, but it is not equivalent to neural-vocoder synthesis and may limit perceived naturalness and detail. A next step is to obtain and validate the compatible HiFi-GAN generator checkpoint and regenerate the samples.

Evaluation depth

The archive contains generated audio and model-prediction plots, but no formal subjective listening study, MOS, intelligibility test or pronunciation annotation. A future evaluation should include native-speaker review, targeted pronunciation checks, and objective measures appropriate to TTS, with clear protocols.

Data documentation

The English split audit is explicit in the project facts. The Telugu archive would benefit from a similarly complete machine-readable audit summarising source recordings, selected durations, segment counts, split overlap and speaker distribution. Such an audit would strengthen reproducibility.

Generalisation

The English workflow records a single speaker label in the training setup, so speaker-independent generalisation should not be claimed. Further work could broaden speaker coverage, evaluate out-of-domain text, refine language-specific normalisation and pronunciation handling, and compare vocoder options.

12. Reproducibility and execution

Dependencies

The repository requirements include PyTorch 2.0 or later, NumPy, SciPy, librosa 0.10 or later, SoundFile, PyWorld, Matplotlib, PyYAML, num2words, g2p_en, NLTK, tqdm and ReportLab. Montreal Forced Aligner is installed separately in a Conda environment rather than through the Python requirements file. Exact working versions should be recorded from the execution environment when preparing a reproducibility package.

Execution approach

The README documents language-specific preprocessing stages and training/inference entry points. The YAML files contain absolute Google Drive paths from the original Colab environment, so users must update those paths to match their own workspace before running. The English workflow includes Svarah manifest preparation and MFA alignment; the Telugu workflow reads the recovered manifests and existing alignment TextGrids.

Checkpoint access

The report-materials ZIP did not include checkpoint binaries. The full submission repository must provide separate English and Telugu checkpoints as required by the assignment. Since the checkpoint files are larger than ordinary GitHub file limits, the public repository should use Git LFS or a persistent external download link and explain the download process in README.md.

Reproducibility command template

```
pip install -r requirements.txt # Update absolute paths in
English/config.yaml and Telugu/config.yaml python
English/preprocess.py --stage prepare python English/preprocess.py --
stage align python English/preprocess.py --stage features python
English/train.py python English/inference.py

python Telugu/preprocess.py --stage prepare
python Telugu/preprocess.py --stage features
python Telugu/train.py python
Telugu/inference.py
```

These commands are templates from the repository workflow; environment paths and data availability must be configured first. Telugu preparation reads existing alignment data as documented in README.md.

13. Conclusion

Summary

The project implements two separate FastSpeech 2-based TTS pipelines for English and Telugu around a shared PyTorch codebase, language-specific text processing and separate configurations. The archived evidence includes training logs, loss plots, ten generated WAV files, inference summaries and all thirty duration, pitch and energy plots. The work addresses the assignment's principal implementation and artifact requirements. The report also records the important synthesis constraint that Griffin–Lim was used because the HiFi-GAN checkpoint was unavailable, and identifies concrete next steps for stronger perceptual evaluation and reproducibility.

References and source materials

1. Ren, Y. et al. “FastSpeech 2: Fast and High-Quality End-to-End Text to Speech.” The project README identifiesthe FastSpeech 2 paper as the architecture reference. Full bibliographic metadata should be verified against the original paper before external publication.

2. IIIT Hyderabad, Language Technologies Research Centre. “Implementation of FastSpeech 2-Based Text-to-SpeechSynthesis” assignment brief, deadline 07 November 2026. Supplied assignment PDF, pages 1–3.

3. AI4Bharat. Svarah speech corpus. Dataset used for the English workflow; cite the dataset's official publication orrepository in the final public version.

4. IndicTTS Telugu. Telugu speech corpus and alignment resources used by the Telugu workflow; cite the dataset'sofficial publication or repository in the final public version.

5. Montreal Forced Aligner documentation and PyWorld documentation, as applicable to the configured Englishalignment and F0 extraction workflow. Exact versioned references should be added from the installed environment.